

# Validation report

This report presents the results of running the StoryFlowDSL EVL validation file against three instance models. The three models tested are:

- StoryDanaModel\_hll.xmi: Dormitory Investigation scenario. Intended valid model (T1).
- StoryDarkRoomModel\_hll.xmi: Frank's Dark Room & Lighthouse scenario. Intended valid model (T1).
- StoryInvalidModel\_hll.xmi: Broken Story. Deliberately constructed to violate constraints (T3).

The EVL file defines 9 constraints across 5 contexts (Story, Chapter, Scene, DiceRule, Effect), covering both structural invariants and semantic domain rules.

## Results

### StoryDanaModel\_hll.xmi

#### Dormitory Investigation

| Constraint | Name                     | Result & Justification                                                                    |
|------------|--------------------------|-------------------------------------------------------------------------------------------|
| C1         | ExactlyOneStartScene     | PASS: corridor_dorm is the only scene with isStart=true.                                  |
| C2         | UniqueChapterOrder       | PASS: single chapter present, no order conflict possible.                                 |
| C3         | ChapterNotEmpty          | PASS: the chapter contains 4 scenes.                                                      |
| C4         | EndingSceneHasNoChoices  | PASS: hallway_exit (isEnding=true) has no outgoing choices.                               |
| C5         | NonEndingSceneHasChoices | PASS: all 3 non-ending scenes have at least one choice.                                   |
| C6         | UniqueSceneld            | PASS: all scene IDs are unique.                                                           |
| C7         | StartCannotBeEnding      | PASS: no scene has both flags set simultaneously.                                         |
| C8         | ValidDiceRange           | PASS: no DiceRule present in this model.                                                  |
| C9         | EffectTargetConsistency  | PASS: SET_VARIABLE effects reference targetVar and ADD_ITEM effects reference targetItem. |

Result: 0 violations. StoryDanaModel\_hll.xmi passes all 9 constraints cleanly.

## StoryDarkRoomModel\_hll.xmi

### Frank's Dark Room

| Constraint | Name                     | Result & Justification                                                                               |
|------------|--------------------------|------------------------------------------------------------------------------------------------------|
| C1         | ExactlyOneStartScene     | PASS: frank_negotiation is the only scene with isStart=true.                                         |
| C2         | UniqueChapterOrder       | PASS: chapters have orders 1 and 2 respectively.                                                     |
| C3         | ChapterNotEmpty          | PASS: both chapters contain scenes.                                                                  |
| C4         | EndingSceneHasNoChoices  | PASS: ending_sacrifice_chloe and ending_save_chloe have no choices.                                  |
| C5         | NonEndingSceneHasChoices | PASS : all 6 non-ending scenes have at least one outgoing choice.                                    |
| C6         | UniqueSceneld            | PASS: all scene IDs are unique across the model.                                                     |
| C7         | StartCannotBeEnding      | PASS: no scene has both flags set simultaneously.                                                    |
| C8         | ValidDiceRange           | PASS: DiceRule: min=1, max=6, threshold=4. All conditions satisfied ( $1 < 6$ , $1 \leq 4 \leq 6$ ). |
| C9         | EffectTargetConsistency  | PASS: ADD_ITEM effects reference targetItem and SET_VARIABLE effects reference targetVar.            |

Result: 0 violations. StoryDarkRoomModel\_hll.xmi passes all 9 constraints cleanly.

## Deliberately Invalid Model

### StoryInvalidModel\_hll.xmi

#### Broken Story

This model was constructed specifically to trigger EVL constraint violations. The model is structurally conformant to the Ecore metamodel but violates domain semantic and structural invariants defined in the EVL file.

| Constraint | Name                     | EVL Error Message                                                                                                     |
|------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------|
| C1         | ExactlyOneStartScene     | Story 'Broken' must have exactly one start scene, but where found: 2                                                  |
| C2         | UniqueChapterOrder       | Story 'Broken' contains two or more chapters with the same order value. Each chapter must have a unique order number. |
| C4         | EndingSceneHasNoChoices  | Scene 'bad_ending' is marked as an ending but has 1 outgoing choice(s). Ending scenes must be terminal.               |
| C5         | NonEndingSceneHasChoices | Scene dead_end is not an ending scene but has no outgoing choices. Add at least one choice.                           |
| C6         | UniqueSceneld            | Duplicate scene id: '4' (two scenes share the same ID value)                                                          |
| C8         | ValidDiceRange           | DiceRule has an invalid configuration: Required: min < max and min <= successThreshold <= max.                        |
| C9         | EffectTargetConsistency  | Effect of type EEnumLiteral [name=SET_VARIABLE, ...] has a missing or inconsistent target.                            |

**Result: 7 violations detected.** Every targeted constraint fired correctly. The model is correctly rejected by EVL before any ETL transformation is attempted.

Validation

- ✗ DiceRule has an invalid configuration: Required: min < max and min <= successThreshold <= max.
- ✗ Duplicate scene id: '4'
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- ✗ Effect of type EEnumLiteral [name=SET\_VARIABLE, value=0, instance=SET\_VARIABLE, literal=SET\_VARIABLE, ] has a missing or inconsistent target.
- ✗ Scene 'bad\_ending' is marked as an ending but has 1 outgoing choice(s). Ending scenes must be terminal.
- ✗ Scene dead\_end is not an ending scene but has no outgoing choices. Add at least one choice.
- ✗ Story 'Broken' contains two or more chapters with the same order value. Each chapter must have a unique order number.
- ✗ Story 'Broken' must have exactly one start scene, but where found:2

## Conclusion

Both valid models (Dana and Dark Room) produce zero violations, confirming they are correct instances of the HLL metamodel. The invalid model produces 7 violations covering 7 distinct constraints, demonstrating that the EVL file reliably detects all targeted classes of domain errors before any transformation is attempted.